

Lawrence Hsiung

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Professional Experience

Prototypes and Concepts Systems Engineer @ Lucid Motors November 2024 - Present

- Design O-1 electrical systems and integrations for 1:1 production experiences on prototype vehicles, special projects, and proof of concepts (e.g. Cosmos, Earth, Lunar).
- Define and build all control schema for HMI UIUX on various simulators and prototype vehicles with Python, Arduino C, Protopie, for future manufacturability and production.
- Build out electrical lab bench validation and integration setup for rapid iteration on solutions for UX and OEM partner teams' design of experiments and test builds.

Product & Systems Engineer @ FarmWise August 2022 - November 2024

- Lead engineering teams on prioritizing and defining the Vulcan smart weeder roadmap and product hardware features for better DFM and maximum customer value.
- Compose and maintain the Vulcan operator's manual and internal technical training documentation leveraging deep technical knowledge of the Vulcan system for safety.

Hardware QA Engineer @ FarmWise Promoted December 2023

- Design and execute comprehensive electrical, mechanical, and hydraulic experiments for the full NPI cycle of the Vulcan smart weeding robot and Intelligent Plant Scanner box.
- Automate analysis of field and lab test data utilizing Python data visualization libraries to efficiently provide constructed metrics for cross functional teams to reduce MTBF.
- Deploy contract manufacturer test fixtures across EVT/DVT/PVT cycles to validate NVIDIA Jetson powered Intelligent Plant Scanners and hydraulic actuators.

Audio Test Engineer @ Zoom August 2020 - September 2021

- Define audio test cases for Zoom's hardware certification programs for key market leaders and own over 100 hardware device certifications with product partners.
- Develop automated Python testing of over 24,000 audio test cases to evaluate new client feature effects on signal quality with VISQOL and ITU-T standard signals.
- Perform root cause analysis and trend tracking on advanced audio issues.

Systems Engineer I @ BAE Systems January 2020 - May 2020

- Analyze sensitive data from field reports on combat vehicles and diagnose fault codes using simulations utilizing hardware in line (HIL) testing in a lab setting.
- Model and simulate electrical system components using proprietary software and build wire diagrams for electrical hardware connections on various combat vehicles.

Electrical Engineering Technician @ Fitbit January 2019 - January 2020

- Test over 50 electrical hardware tests suites for prototype Versa 2, Versa, Inspire, and Inspire HR wearable product DUTs in electrical and mechanical laboratory settings.
- Automate test suites using Python to extract metrics from user released data, providing insights into iterative audio hardware design on prototypes.
- Conduct mass market Versa 2 & 3 microphone port acoustic validation, utilizing MATLAB and Python to analyze sensor and acoustic data.

Education

MSEE @ SJSU January 2019 - June 2022

Double Focus in Machine Learning and Power Electronics

BSEE @ UCSD August 2014 - May 2018

Focus in Machine Learning and Controls, Minor in Business Studies

Skills

Systems & Architecture : O-1 Architecture Design, System Bring-up, Rapid Prototyping

EE Focuses: Power Electronics, Analog/Digital/Mixed Signal, Digital Signal Processing

Software & Firmware: Arduino C, Python, MATLAB, Low-level Scripting, Machine Learning